Evaluation To evaluate general performance, we again use MMLU (Hendrycks et al., 2020). Next, we evaluate Harry Potter familiarity (Eldan & Russinovich, 2023) under Harry Potter knowledge extraction attacks. Full details are available in Appendix G. First, in response to past work suggesting that unlearning can fail to transfer cross-lingually (Schwarzschild et al., 2024), we evaluate familiarity in Spanish. Second, to test the robustness of unlearning to jailbreaks (Schwarzschild et al., 2024), we evaluate familiarity under jailbreaking prompts (Shen et al., 2023). Third and fourth, we evaluate the extent to which the model is robust to knowledge extraction attacks (Lu et al., 2022; Ishibashi & Shimodaira, 2023; Patil et al., 2023; Shi et al., 2023; Schwarzschild et al., 2024) in the form of high-level summaries and short snippets of text from the Harry Potter books.

LAT helps to more robustly unlearn Harry Potter knowledge. We present results in Table 4. WHP-C-LAT Pareto dominates WHP and WHP-C across all measures except MMLU.

4.3.2 Unlearning WMDP Biology and Cyber Knowledge

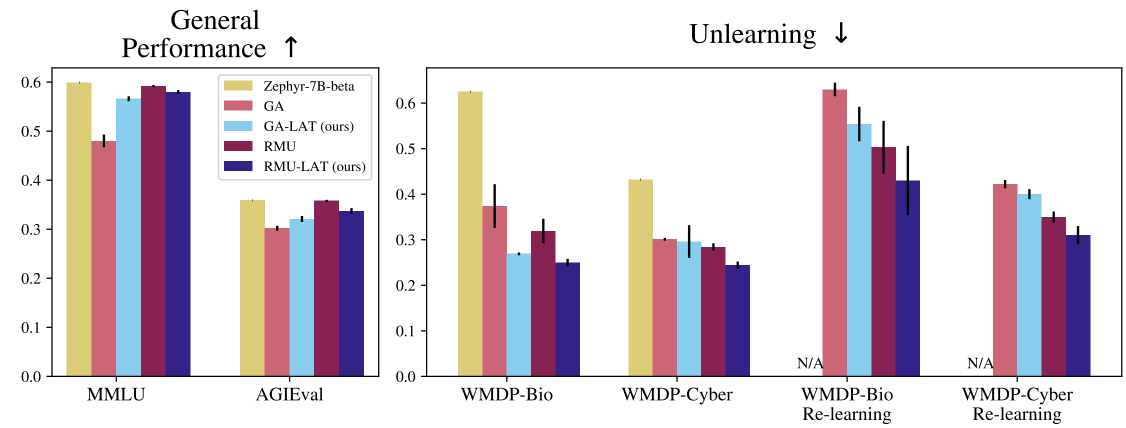
Following Li et al. (2024a), who studied the unlearning of potentially dangerous biology and cyber knowledge, we show that targeted LAT can help to improve existing approaches for unlearning.

Data As in as in Li et al. (2024a), we use the WMDP biology and cyber corpora as *forget* datasests and WikiText (Merity et al., 2016) as a *retain* dataset.

Model and methods As in Li et al. (2024a), we use Zephyr-7B off the shelf (Tunstall et al., 2023). We test two different unlearning methods with and without targeted LAT. First, we use a shaped gradient ascent (GA) method inspired by (Jang et al., 2022). We fine-tune the model to jointly minimize training loss on the retain set and a $\log(1-p)$ loss on the forget set as done in Mazeika et al. (2024). To augment GA with targeted LAT, we apply latent-space perturbations optimized to minimize training loss on the forget set. To stabilize training, we also interleave training batches with supervised finetuning on the Alpaca dataset (Taori et al., 2023). Second, we use representation misdirection for unlearning (RMU) from Li et al. (2024a). With RMU, the model is trained at a given layer to (1) map activations from forget-set prompts to a randomly sampled vector while (2) leaving activations from other prompts unaltered. To augment RMU with targeted LAT, we apply latent-space adversarial perturbations only when training on the forget set. We optimize these perturbations to minimize the model's cross-entropy training loss on the undesirable forget-set example. We experimented with various layer combinations and found the best results from applying them to the activations immediately preceding the RMU layer.

Evaluation We evaluate how well the model's general capabilities have been preserved by testing on MMLU (Hendrycks et al., 2020) and AGIEval (Zhong et al., 2023). We evaluate the effectiveness of unlearning in the model using biology and cyber knowledge assessments from Li et al. (2024a). These multiple choice evaluations represent a qualitatively different task than the forget sets (which were full of bio and cyber documents), so they test the ability of LAT to generalize to qualitatively different kinds of unwanted behaviors than those used during fine-tuning. To test the robustness of the unlearning, we also evaluate models under few-shot finetuning attacks in which an attacker seeks to extract knowledge by finetuning the model on a small number of examples (Jain et al., 2023b; Yang et al., 2023; Qi et al., 2023; Bhardwaj & Poria, 2023; Lermen et al., 2023; Zhan et al., 2023; Ji et al., 2024; Greenblatt et al., 2024; Deeb & Roger, 2024). Here, we use a simple but surprisingly effective attack: we randomly sample a single batch of 2 examples from the relevant forget set and repeatedly train on that single batch for 20 iterations. We then report the highest WMDP bio/cyber performances for each model across evaluation checkpoints at 5, 10, and 20 steps. For all evaluations, we use 1,000 samples on lm-evaluation-harness v0.4.0 Gao et al. (2023) as done in Li et al. (2024a).

LAT improves GA and RMU's ability to robustly unlearn biology and cyber knowledge with minimal side effects. Table 5 shows results for evaluating models by MMLU versus unlearning effectiveness. GA-LAT outperforms GA by a large margin under all evaluations. Similarly, RMU-LAT outperforms RMU in all evaluations, except for a 1.2% decrease in MMLU and 2.1% decrease in AGIEval. Across all



Model	General Performance ↑		Unlearning ↓		Unlearning + Re-learning ↓	
	MMLU	AGIEval	WMDP-Bio	WMDP-Cyber	WMDP-Bio	WMDP-Cyber
Zephyr-7B-beta	0.599	0.395	0.625	0.432	-	-
GA	0.480$\pm$0.013	0.302$\pm$0.005	0.374$\pm$0.048	0.301$\pm$0.003	0.630$\pm$0.015	0.422$\pm$0.009
GA-LAT (ours)	**0.566**$\pm$0.005	**0.321**$\pm$0.06	**0.269**$\pm$0.03	**0.296**$\pm$0.036	**0.554**$\pm$0.038	**0.400**$\pm$0.011
RMU	**0.592**$\pm$0.002	**0.358**$\pm$0.002	0.319$\pm$0.027	0.284$\pm$0.008	0.503$\pm$0.058	0.350$\pm$0.012
RMU-LAT (ours)	0.580$\pm$0.004	0.337$\pm$0.006	**0.250**$\pm$0.008	**0.244**$\pm$0.008	**0.430**$\pm$0.074	**0.310**$\pm$0.020

Table 5: **LAT can improve gradient ascent (GA) and representation misdirection for unlearning (RMU)'s ability to unlearn the WMDP biology and cyber datasets (Li et al., 2024a) with minimal side effects**. We evaluate models' general performance using MMLU and AGIEval and its unlearning with the WMDP bio and cyber evaluations from Li et al. (2024a). The random-guess baseline for WMDP bio/cyber is 25%. Finally, to evaluate robustness to re-learning, we report WMDP performance after up to 20 iterations of repeatedly retraining on a single batch of 2 examples. We report means and standard error of the means over $n = 3$ runs with different random seeds.

experiments, it is surprisingly easy for the unlearned models to re-learn the unwanted knowledge. Repeatedly training on the same batch of 2 examples for up to 20 iterations improved WMDP bio/cyber performance by an average of 15.7 percentage points. However, LAT makes the models more resistant to re-learning. On average, re-learning closed 74.7% of the performance gap between the unlearned model and the original model for non-LAT methods but only 59.9% of the gap for LAT methods.

5 Discussion

LAT can effectively augment existing state-of-the-art fine-tuning and adversarial training methods. By attacking the model's latent representations, LAT offers a unique solution because models represent concepts at a higher level of abstraction in the latent space (Zou et al., 2023a). Here, we have used targeted latent adversarial training (LAT) to strengthen existing defenses against persistent harmful behaviors in LLMs. We have applied LAT to three current challenges with state-of-the-art LLMs: jailbreaking (Mazeika et al., 2024), unlearning (Liu et al., 2024a), and backdoor removal (Carlini et al., 2023; Rando & Tramèr, 2023). In each case, we have shown that LAT can augment existing techniques to improve the removal of unwanted behaviors with little or no tradeoff in general performance. Overall, these results support but do not yet confirm our hypothesis that LAT can remove neural circuitry from models responsible for undesirable behaviors. We leave analysis of the mechanisms behind harmful model behaviors (e.g., (Arditi et al., 2024)) to future work.

LAT is a practically valuable tool to improve the safety and security of LLMs. Our motivation for LAT is a response to two observations. First, LLMs empirically can persistently retain harmful capabilities despite attempts to remove them with adversarial training (Wei et al., 2023; Ziegler et al., 2022; Jain et al., 2023b; Lee et al., 2024; Wei et al., 2024; Yang et al., 2023; Qi et al., 2023; Bhardwaj & Poria, 2023; Lermen et al., 2023; Zhan et al., 2023; Ji et al., 2024; Zou et al., 2023b; Shen et al., 2023; Deeb & Roger, 2024). Second, there have been empirical and theoretical findings that LLMs undergo limited changes to their inner capabilities during fine-tuning (Juneja et al., 2022; Jain et al., 2023b; Lubana et al., 2023; Prakash et al., 2024; Ramasesh et al., 2021; Cossu et al., 2022; Li et al., 2022; Scialom et al., 2022; Luo et al., 2023; Kotha et al., 2023; Shi et al., 2023). All three problems that we have used targeted LAT to address – jailbreaks, backdoors, and undesirable knowledge – are ones in which an LLM exhibits harmful behaviors that are difficult to thoroughly remove. Our results show that targeted LAT can be useful for making models more robust to these persistent failures. We also find that these failure modes need not be precisely known for LAT to be helpful, showing instances in which LAT can improve generalization to different datasets of attack targets, harmful behaviors, and knowledge-elicitation methods than were used during training.

LLM unlearning techniques are surprisingly brittle. In Section 4.3, we find that state-of-the-art LLM unlearning methods are surprisingly vulnerable to relearning from small amounts of data. We find that re-training repeatedly on only *two* samples from the forget set was consistently able to close more than half of the performance gap between the original and unlearned models on average. We find that targeted LAT can reduce the sample efficiency of re-learning, but there is much room for improvement in designing unlearning methods that are robust to few-shot finetuning attacks. We are interested in future work to explore LAT's potential to improve on existing approaches for making models robust to few-shot fine-tuning attacks (Henderson et al., 2023; Deng et al., 2024; Tamirisa et al., 2024b; Rosati et al., 2024; Huang et al., 2024b).

Limitations – attack methodology and model scale. While we have shown that LAT can be useful, it can also be challenging to configure and tune. In our experience, we found the selection of dataset, layer(s), and perturbation size, to be influential. We also found that interleaving supervised finetuning in with training and NaN handling were key to stable training. LAT can be done in different layers, with various parameterizations, and under different constraints. Our work here is limited to residual stream perturbations designed with projected gradient descent. Additionally, all of our experiments are done in LLMs with fewer than 10 billion parameters. Due to LAT's usefulness across model families and training algorithms, we expect that this usefulness will extend to larger models. However, future work will be needed to confirm this.

Future work

- **Improved latent-space attacks** In addition to performing LAT with perturbations to an LLM's residual stream, we are interested in other strategies for attacking its internal representations. Toward this goal, engaging with recent work on LLM representation engineering and interpretability may help to better parameterize and shape latent space attacks. Specifically, we are interested in studying LAT with an adversary parameterized by perturbations to a Sparse Autoencoder's encodings (Cunningham et al., 2023) or the weights of low-rank adapters (Zou et al., 2023a; Wu et al., 2024). We also speculate that universal attacks instead of single-instance attacks may be more interpretable and might better target the most prominent mechanisms that a model uses when it produces undesirable outputs.

- **Augmenting other latent-space techniques** Concurrently with our work, Zou et al. (2024), Rosati et al. (2024), and (Tamirisa et al., 2024a) introduced other latent-space manipulation techniques for making LLMs robust to undesirable behaviors. We are interested in studying how these techniques compare to LAT and whether LAT can be used to improve them.

- **Generalized adversarial attacks for LLM evaluations** We are interested in the extent to which embedding-space attacks (e.g., (Schwinn et al., 2023)), latent-space attacks, (e.g., (Casper et al., 2024b)), and few-shot fine-tuning attacks (e.g., (Qi et al., 2023)) can improve evaluations of LLM safety (Casper et al., 2024a).